How many Hamiltonian cycles are there in $K_{10,10}$?

Asked 7 years, 6 months ago Modified 7 years, 6 months ago Viewed 394 times

0

I want to calculate the number of Hamiltonian cycles in $K_{10,10}$. Could anyone help me? I think in K_{10} we have $9!$ Hamiltonian cycles.

discrete-mathematics graph-theory computer-science trees hamiltonian-path

1

Share Cite Edit Follow Flag

edited Mar 13, 2015 at 14:43
anomaly
23.3k 5 37 77

asked Mar 12, 2015 at 17:06
Prof. Kosi Noura
25 8

1 I think in K_{10} we have $9!/2$ Hamiltonian cycles. Tell me, how many Hamiltonian cycles do you have in K_3? – bof Mar 12, 2015 at 18:39

@bof, i have no idea :) – Prof. Kosi Noura Mar 12, 2015 at 18:42

3 You think you know the answer for K_{10} but you have no idea about K_3? – bof Mar 12, 2015 at 18:45

@bof, i search on google. – Prof. Kosi Noura Mar 12, 2015 at 18:51

Does this answer your question? The number of Hamiltonian cycles in the complete bipartite graph – Andrew Uzzell Apr 8 at 16:11

2 Answers

Sorted by: Highest score (default)

0

For all n≥3, the number of distinct Hamilton cycles in the complete graph K_n is (n−1)!/2.

So for K_{10}, $9!/2$

Share Cite Edit Follow Flag

answered Mar 12, 2015 at 20:28
Ryan
63 5

1 The question is about the complete bipartite graph $K_{10,10}$, not the complete graph K_{10}. – Casteels Mar 13, 2015 at 12:21

Are you asking about complete *bipartite* graphs? If yes, then please look here.

0

Share Cite Edit Follow Flag

answered Mar 13, 2015 at 0:43



HEKTO
1,258 10 14

would u please add more detail ? – Prof. Kosi Noura Mar 13, 2015 at 6:28

General formula for the number of (directed) Hamiltonian cycles for $K_{n,n}$ is $n!(n-1)!$ – HEKTO Mar 13, 2015 at 14:35